FE1101

Introduction to Food Business and Development

FULL MARKING SCHEME

Examination Session	Summer 2024
Paper Number	1
Duration	1.5 hours
Instructions to Candidates	Answer one question from Section One and one question from Section 2. All questions carry equal marks. Use separate answer books for each section.
Sections Covered	Section One — Co-operatives · Section Two — Development
Internal Examiners	Dr Bridget Carroll · Dr Stephen Onakuse · Dr Emma Beacom

Model answers, mark allocations and examiner notes for all four questions across both sections, so revision is complete regardless of which pair you are asked to answer.

SECTION ONE — CO-OPERATIVES

Question 1

50% of paper

QUESTION AS SET

Explain how co-operatives help farmers improve the effectiveness and profitability of their individual businesses.

FULL MODEL ANSWER

A producer co-operative's core aim is to improve the individual member's own business — the co-operative itself is a vehicle, not the end goal. It does this through six distinct mechanisms, each of which addresses a specific limitation an individual farmer faces trading alone.

1. Countervailing power against concentrated buyers and input suppliers

A single farmer typically faces a market with relatively few buyers for their output and few suppliers of key inputs — a structurally weak bargaining position. By pooling members' output and input needs, a co-operative gives farmers countervailing power in these negotiations, allowing them to secure better prices on both sides of their business rather than accepting whatever an individual buyer or supplier offers.

2. Economies of scale in processing and marketing

Individual farmers rarely have the scale to run their own processing plant or export operation profitably. Co-operatives coordinate processing and marketing across the whole membership, capturing economies of scale that directly lower the per-unit cost of getting a member's product to market and improve the price it ultimately fetches.

3. Access to credit, markets and services an individual could not secure alone

A co-operative can access credit facilities, distribution channels and international markets that would be closed to an individual smallholder trading alone — because a co-operative's larger scale and pooled asset base make it a more viable counterparty for lenders and international buyers.

4. Risk-sharing and income stabilisation

Pooling output and, in some cases, pooling exposure to price volatility gives individual members a more stable income than they would achieve exposed individually to market swings — directly improving the predictability, and often the level, of an individual farm business's profitability.

5. Vertical integration — capturing profit from further up the chain

Where a co-operative moves into processing, branding or export marketing, it captures value that would otherwise accrue to an intermediary further along the supply chain, and returns that value to members. **Ornua** illustrates this directly: owned by Irish dairy producer co-operatives, Ornua markets roughly 60% of Irish dairy exports internationally under the Kerrygold brand, generating approximately €3.5 billion in exports across more than 100 markets (Bord Bia, 2021). The branding and export-market value Kerrygold captures internationally flows back to the individual dairy farmers who supply the primary co-operatives that own Ornua — value an individual farmer selling raw milk could never capture alone.

6. Coordination, standardisation and technical assistance

Co-operatives act as coordinators of quality assurance and provide technical assistance, better inputs and standardisation across the membership — improving the underlying effectiveness of each member's farming operation, not just the price they receive for what they produce.

> **Sourcing note.** *All six mechanisms and the Ornua/Kerrygold figures are drawn directly from the Section 2 (Producer Co-operatives) lecture material, including the Bord Bia (2021) export figure cited on the same slide.*

Worked example: a small, newly formed co-operative

Ring of Kerry Quality Lamb demonstrates these mechanisms at a much smaller, more recent scale: thirty lamb producers came together specifically to maximise the price received for their lamb, selling directly to hotels and restaurants rather than through a conventional marketing chain, with quality certification used to add value and diversify beyond the standard commodity market. This is mechanism 1 (better terms through collective selling) and mechanism 5 (capturing value further up the chain, from farm-gate price to a certified, branded product sold direct to food service) operating together, at a scale an individual sheep farmer could not replicate alone.

Conclusion: across all six mechanisms, the common thread is that a co-operative solves a specific limitation of operating as a single, small, individual business — weak bargaining power, insufficient scale, restricted market access, concentrated risk, and value leaking to intermediaries — without requiring the individual farmer to give up ownership of their own farm business. The co-operative model directly targets effectiveness (better inputs, coordination, standardisation) and profitability (better prices, captured value, income stability) simultaneously, which is precisely why it has remained the dominant structure in Irish and international agriculture.

Mark Allocation

Band	Range	Criteria
First Class	70–100%	Explains at least four distinct mechanisms (bargaining power, economies of scale, market/credit access, risk-sharing, vertical integration, coordination/standardisation) with correct reasoning for each, AND uses at least one concrete example (Ornua/Kerrygold or Ring of Kerry Quality Lamb) to demonstrate the mechanism in practice, AND explicitly distinguishes effectiveness gains from profitability gains rather than treating them as the same thing.
Upper Second (2.1)	60–69%	Three to four mechanisms explained correctly with reasonable detail, but examples are generic or absent, or the effectiveness/profitability distinction is not made explicit.
Second Class / Pass	40–59%	One or two mechanisms described in general terms (e.g. "better prices") without explaining the underlying reason why the co-operative structure delivers this.
Fail	0–39%	Generic description of co-operatives with no clear mechanism linking co-operative membership to an individual farm business's effectiveness or profitability.

Question 1 total — 100% (50% of overall paper)

Question 2

50% of paper

QUESTION AS SET

Outline and discuss the distinctive characteristics of co-operatives using examples to support your answer.

FULL MODEL ANSWER

Characteristic 1 — Unified ownership, control and use

In a co-operative, the same group of people owns the business, controls it, and uses its services — unlike a conventional company, where owners (investors) and users (customers) are typically different groups. A co-operative is, by definition, a member-owned business.

Characteristic 2 — Democratic control on a one-member-one-vote basis

Control is exercised democratically at the AGM and through elected committees, with each member holding one vote regardless of how much capital they have contributed — not one-share-one-vote as in a conventional company. This is a structural, not cultural, difference: it is written into how the co-operative is legally constituted.

Characteristic 3 — Surplus distributed by use, not by capital

Members contribute capital but usually receive only limited compensation on it; the co-operative's surplus is instead distributed largely according to each member's *use* of the co-operative (the patronage refund) — a fundamentally different reward logic to a conventional shareholder dividend.

Characteristic 4 — A distinctive multi-tier federated structure

Irish agricultural co-operatives are organised across genuinely distinct tiers, which has no direct equivalent in a typical single-company corporate structure:

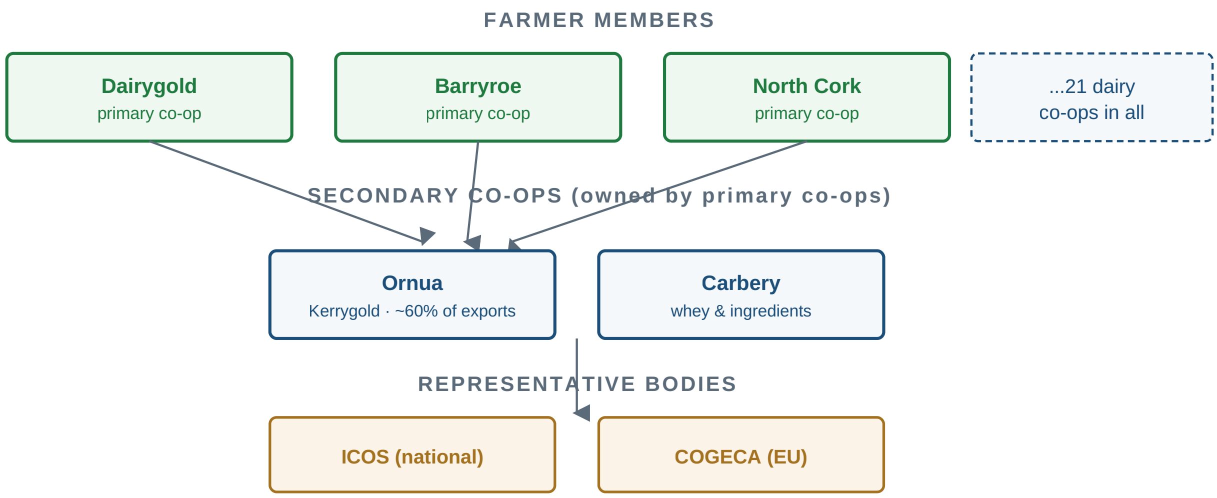

Figure 1. The tiered structure of Irish dairy co-operatives — individual farmer members up through primary co-ops, secondary co-ops, and national/EU representative bodies.

- **Primary co-ops** — made up of individual farmer members, e.g. Dairygold, Barryroe, North Cork Co-op.
- **Secondary co-ops (federations)** — made up of individual co-ops rather than individual farmers, e.g. Ornua (dairy marketing) and Carbery (whey processing).
- **Representative bodies** — ICOS nationally, COGECA at EU level, representing co-operative interests in policy rather than trading directly.

This tiering lets small primary co-ops achieve the scale of a large multinational (through their shared secondary co-op) while each individual primary co-op remains democratically owned by its own local farmer members — a structure a conventional single-tier corporate subsidiary model does not replicate.

Characteristic 5 — Purpose: improving the member's business, not maximising co-op profit as an end in itself

A producer co-operative's stated aim is to improve the member's own business and the wider industry — the prime beneficiary is the individual producer, not the co-operative as a standalone profit-seeking entity. This differs from a conventional company, whose primary purpose is return to its shareholders as an end in itself.

Characteristic 6 — Concern for community beyond the immediate membership

The seventh co-operative principle directs co-operatives toward sustainable development of their local communities — sponsorship, environmental protection, job creation — extending the co-operative's obligations beyond its direct membership in a way a conventional business, focused on shareholder return, is not structurally obliged to do.

> **Sourcing note.** *The primary/secondary/representative-body tier structure and the named examples (Dairygold, Barryroe, North Cork, Ornua, Carbery, ICOS, COGECA) are taken directly from the Section 2 lecture slides on Irish dairy co-operatives. Ornua's export share and value are the same Bord Bia (2021)-sourced figures used on the same slide set.*

Illustrative example — Barryroe Co-op and Carbery: this pairing shows Characteristics 1–4 operating together in practice — Barryroe is a primary co-op, democratically owned and controlled by its individual farmer members (Characteristics 1–3), which in turn is a part-owner of Carbery, a secondary co-op processing whey at a scale no individual primary co-op could achieve alone (Characteristic 4).

Mark Allocation

Band	Range	Criteria
First Class	70–100%	Outlines at least four distinct characteristics (ownership/control/use unity, democratic control, patronage-based surplus, multi-tier structure, member-focused purpose, community obligation) AND discusses each rather than just listing it AND supports the answer with at least one real, correctly-used example (Dairygold/Barryroe/Ornua/Carbery/ICOS).
Upper Second (2.1)	60–69%	Three to four characteristics outlined correctly with some discussion, but examples are generic, missing, or used only as a label rather than to illustrate the characteristic.
Second Class / Pass	40–59%	Two or fewer characteristics identified, largely listed rather than discussed, with weak or no supporting examples.
Fail	0–39%	General description of what a co-operative is with no clear characteristics identified and no real-world examples used.

Question 2 total — 100% (50% of overall paper)

SECTION TWO — DEVELOPMENT

Section Two, Question 1

50% of paper

QUESTION AS SET

Tackling inequality is central to the Sustainable Development Goals (SDGs). Enumerate and discuss one of the challenges relevant to the SDGs.

FULL MODEL ANSWER

Enumeration — challenges relevant to achieving the SDGs

- **Global shocks reversing progress** — e.g. the COVID-19 pandemic, discussed in depth below.
- **The financing / means-of-implementation gap** — SDG 17 exists precisely because achieving the other 16 goals requires a scale of financing, technology transfer and capacity-building that many countries cannot mobilise alone.
- **Data and monitoring gaps** — tracking progress against 169 targets requires reliable, disaggregated national data that many low- and medium-income countries do not yet consistently produce.
- **Competing national priorities** — because the goals are universal but national governments set their own domestic priorities, short-term political and economic pressures can crowd out longer-term SDG commitments.

The challenge discussed in depth below is the first: the disruptive impact of a major global shock — specifically COVID-19 — on SDG progress, because it connects most directly and most measurably to the inequality theme in the question.

Discussion — COVID-19 as a setback to the SDGs, and to inequality specifically

The COVID-19 pandemic represents the clearest recent illustration of how quickly a global shock can undo SDG progress and widen the very inequality the goals are meant to close. According to UNDP's own analysis (Abidoye, Felix, Kapto & Patterson, 2021, *Leaving No One Behind: Impact of COVID-19 on the SDGs*), the pandemic's combined health, education and living-standards impact caused what may be the first-ever decline in the Human Development Index since its creation — equivalent to erasing roughly six years of prior human development progress.

The scale of the setback to poverty specifically is stark: without a change in policy effort, the number of people in extreme poverty in low- and medium-development countries was projected to rise as high as 813 million because of COVID-19, before gradually declining to a projected 465.5 million (37.5% of the population) by 2030 in low human development countries, and 161 million (6.4% of the population) in medium human development countries. Crucially, this impact was not distributed evenly — it fell hardest on countries and populations that were already the most vulnerable, directly widening the gap the SDGs, and SDG 10 in particular, are designed to close.

UNDP's modelling also points to the way out: under an integrated "SDG Push" scenario — combining targeted investment in governance, social protection, the green economy and digitalisation — countries can recover lost ground significantly faster than under a "COVID Baseline" of business-as-usual. For example, Namibia's projected timeline for reducing extreme poverty to 4% moves from 2050 under the COVID Baseline to 2035 under an SDG

Push — a gain of 15 years — and more than ten countries are estimated to accelerate their extreme-poverty targets by more than fifteen years under the same integrated approach.

> **Sourcing note.** *All figures in this section (the HDI reversal, the 813 million / 465.5 million / 161 million poverty projections, and the Namibia/Madagascar/Kenya/Laos country examples) are drawn from the UNDP and Frederick S. Pardee Center flagship publication provided in the module's reading list, and independently verified as a genuine 2021 UNDP publication.*

Why this specific challenge matters for the inequality theme: the COVID-19 case demonstrates that tackling inequality is not simply a matter of setting a static 2030 target and progressing steadily toward it — the pandemic shows that inequality-reduction gains can be rapidly reversed by a shock that hits the most vulnerable hardest, and that recovering from such a shock requires the same kind of integrated, cross-goal policy response (health, social protection, economy, technology together) that Integration, as an underpinning element of the SDG framework, was designed to demand. A single-sector response would not have been sufficient to recover the ground COVID-19 took away.

Mark Allocation

Band	Range	Criteria
First Class	70–100%	Enumerates multiple genuine challenges (not just restating "poverty" or "inequality" as challenges), THEN develops one in real depth with accurate supporting evidence, AND explicitly explains why that specific challenge is relevant to the inequality theme in the question stem.
Upper Second (2.1)	60–69%	A reasonable list of challenges given, with one discussed correctly, but the connection back to inequality specifically is asserted rather than explained.
Second Class / Pass	40–59%	One challenge discussed in general terms without evidence, or the enumeration step is missing/very thin.
Fail	0–39%	No clearly identified challenge, or the answer restates SDG content generally without addressing "challenges" as the question asks.

Section Two, Question 1 total — 100% (50% of overall paper)

Section Two, Question 2

50% of paper

QUESTION AS SET

Enumerate and discuss two key elements underpinning the Sustainable Development Goals (SDGs).

FULL MODEL ANSWER

Enumeration

The SDGs are a set of 17 goals for the world's future to 2030, backed by 169 detailed targets, negotiated over two years at the United Nations and agreed by nearly all the world's nations on 25 September 2015. What is new and different about the SDGs, relative to the Millennium Development Goals that preceded them, is captured in three elements: **Universality**, **Integration**, and **Transformation**. Two are developed in depth below.

Element 1 — Universality

Unlike the Millennium Development Goals, which were framed largely as targets for developing countries to meet with donor support, the SDGs apply to every nation and every sector — cities, businesses, schools and organisations are all challenged to act, not just low-income states. This is a genuine structural break from the MDG model: a high-income country is just as much a subject of the SDG framework as a low-income one, which is what allows goals like SDG 10 (reduced inequality) to hold wealthy nations accountable for domestic inequality, not only for foreign aid contributions.

Element 2 — Integration

The 17 goals are explicitly designed as indivisible and interconnected: progress, or failure, on one goal affects the others. This is a deliberate design choice, not an incidental observation — it means, for example, that progress on SDG 1 (no poverty) cannot be sustained without simultaneous progress on SDG 8 (decent work), SDG 10 (reduced inequality) and SDG 13 (climate action). A government that pursues one goal in isolation, ignoring its effect on the others, works against the framework's own design logic.

Sourcing note. *The 17 goals / 169 targets / 25 September 2015 adoption date, and the Universality · Integration · Transformation framing, are taken directly from the Dr Stephen Onakuse SDG lecture material provided for the module.*

A note on the third element (Transformation)

A candidate could equally choose Transformation as one of their two elements: the SDGs are framed not as incremental improvement but as a structural shift in how economies and societies operate, since goals of this scale and 2030 deadline cannot be met through marginal policy adjustment alone. Examiners should credit any two of the three elements, provided each is correctly explained in its own right — unlike the equivalent question on the Summer 2025 paper, this version does not require the discussion to be anchored to inequality specifically, so a correct, well-developed explanation of any two elements on their own terms should receive full credit.

Mark Allocation

Band	Range	Criteria
First Class	70–100%	Correctly enumerates the three-element framework and then discusses (not just names) two of the three — Universality, Integration, Transformation — with accurate supporting SDG facts (17 goals, 169 targets, 2015 adoption) and a clear explanation of what is genuinely new or distinctive about each relative to prior development frameworks.
Upper Second (2.1)	60–69%	Two elements correctly identified and explained, but the explanation is descriptive rather than analytical (i.e. states what each element is without explaining why it represents a genuine break from previous frameworks).
Second Class / Pass	40–59%	One element explained in reasonable depth, or two elements named with only superficial explanation; some factual imprecision about the SDG framework.
Fail	0–39%	SDGs described in general terms with no reference to the specific "elements" framework taught, or significant factual errors about the framework.

Section Two, Question 2 total — 100% (50% of overall paper)